Limits in Finance

From discrete models to continuous risk optimization — the complete mathematics behind the project

The question. A saver can keep money in a risk-free account paying $r = 3\%$ per year, or invest in a stock index with expected return $\mu = 10\%$ and volatility $\sigma = 20\%$. **What fraction f of the savings should go into the stock** so that wealth grows as fast as possible, while respecting the rule *"with 95% confidence, do not lose more than 15% in one year"*?

The answer (derived in Part V): $f_{opt} = 0.705$ — put **70.5% in the stock and 29.5% risk-free**. If rare crashes can happen (Part VI): 61.7%. With gold and bonds available too (Part VII): about **73% stocks**, the rest mostly bonds.

The method. Real finance is *discrete* (interest paid monthly, prices quoted tick by tick, portfolios rebalanced daily). We let the step size go to zero — a **limit** — and obtain continuous models with closed-form expressions. Then **continuity** (the Intermediate Value Theorem) guarantees the optimal allocation exists, and bisection computes it.

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Parameters used throughout.

Module	Parameters
1 Compounding	$P = 10{,}000$, $r = 10\%$, $t = 10$ years
2 Options	$S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$
3–4 Market	$\mu = 10\%$, $\sigma = 20\%$, $r = 3\%$, $T = 1$ year, $\alpha = 95\%$, loss limit $L = 15\%$, leverage cap $f_{max} = 2$
4b Crashes	rate $\lambda = 0.1$ per year, size $J = -25\%$
5 More assets	gold $\mu = \ln 1.05$, $\sigma = 15\%$; bonds $\mu = \ln 1.045$, $\sigma = 6\%$; correlations 0 (stocks–gold), −0.2 (stocks–bonds), 0.3 (gold–bonds); no borrowing

All rates are continuously compounded (Module 1): $\mu = 10\%$ means an average gain of $e^{0.1} - 1 = 10.5\%$ a year, $r = 3\%$ means 3.05%.

Part I — Mathematical toolkit

This part collects every theorem the project relies on, so that later parts can quote them.

1.1 Limits of sequences and continuity

Definition 1.1 (limit). A sequence (a_n) converges to a, written $a_n \to a$, if for every $\varepsilon > 0$ there is N such that $|a_n - a| < \varepsilon$ for all $n \geq N$.

Definition 1.2 (continuity). h is continuous at x if for every $\varepsilon > 0$ there is $\delta > 0$ with $|y - x| < \delta \Rightarrow |h(y) - h(x)| < \varepsilon$.

Lemma 1.3 (sequential continuity). h is continuous at x if and only if $x_n \to x$ implies $h(x_n) \to h(x)$. In words: **a limit can be moved inside a continuous function**,

$$\lim_{n\to\infty} h(x_n) = h\Big(\lim_{n\to\infty} x_n\Big).$$

We use Lemma 1.3 constantly, typically with $h = \exp$: to find $\lim e^{x_n}$ it suffices to find $\lim x_n$.

Theorem 1.4 (squeeze; monotone convergence). (a) If $b_n \leq a_n \leq c_n$ and $b_n, c_n \to a$, then $a_n \to a$. (b) An increasing sequence that is bounded above converges (completeness of $\mathbb{R}$).

1.2 The exponential and the logarithm

Lemma 1.5 (logarithm bounds). For every $u > -1$, $u \neq 0$:

$$\frac{u}{1+u} \;<\; \ln(1+u) \;<\; u. \tag{1}$$

Proof. Let $\psi(u) = u - \ln(1+u)$. Then $\psi(0) = 0$ and $\psi'(u) = 1 - \frac{1}{1+u} = \frac{u}{1+u}$, which is negative on $(-1, 0)$ and positive on $(0, \infty)$; so ψ has a strict minimum at 0 and $\psi(u) > 0$ for $u \neq 0$. That is the right inequality. For the left one apply the right inequality to $v = -u/(1+u) > -1$: $\ln(1+v) = -\ln(1+u) < v = -u/(1+u)$. ∎

Lemma 1.6 (Taylor expansion). For $|u| < 1$,

$$\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots = u - \frac{u^2}{2} + O(u^3), \qquad e^u = 1 + u + \frac{u^2}{2} + O(u^3).$$

Notation: $O(u^k)$ means "bounded by a constant times $|u|^k$ for small u".

1.3 The derivative is a limit

Definition 1.7.

$$h'(x) = \lim_{\Delta \to 0} \frac{h(x+\Delta) - h(x)}{\Delta}.$$

The fraction is the *difference quotient* (slope of a secant line); its limit is the slope of the tangent.

Second-derivative test / concavity. If $h''(x) < 0$ for all x, then h is strictly concave, h' is strictly decreasing, and any point with $h'(x^*) = 0$ is the unique global maximum.

1.4 The Intermediate Value Theorem and the Extreme Value Theorem

Theorem 1.8 (Intermediate Value Theorem, IVT). Let $h : [a,b] \to \mathbb{R}$ be continuous with $h(a) < 0 < h(b)$ (or the reverse). Then there exists $c \in (a,b)$ with $h(c) = 0$.

Proof. Let $c = \sup\{x \in [a,b] : h(x) \le 0\}$, which exists by completeness. If $h(c) < 0$, continuity gives a neighbourhood to the right of c where $h < 0$, contradicting the definition of sup. If $h(c) > 0$, continuity gives a neighbourhood of c where $h > 0$, so a smaller number would be an upper bound, again a contradiction. Hence $h(c) = 0$. ∎

Geometric meaning: a continuous graph cannot get from below the axis to above it without crossing it. Continuity is essential — a function with a jump can skip over 0.

Theorem 1.9 (Extreme Value Theorem). A continuous function on a closed bounded interval $[a,b]$ attains a maximum and a minimum.

1.5 Bisection — the IVT as an algorithm

Algorithm. Start with $[a_0, b_0] = [a,b]$, $h(a)h(b) < 0$. For $k = 0, 1, 2, \ldots$: let $m_k = (a_k + b_k)/2$. If $h(a_k)h(m_k) < 0$ set $[a_{k+1}, b_{k+1}] = [a_k, m_k]$, otherwise $[a_{k+1}, b_{k+1}] = [m_k, b_k]$.

Theorem 1.10 (convergence of bisection). If h is continuous on $[a,b]$ with $h(a)h(b) < 0$, the midpoints m_k converge to a root c of h, and

$$|m_k - c| \le \frac{b-a}{2^{k+1}}.$$

Hence to reach accuracy tol it suffices to take $k \ge \log_2\big((b-a)/\mathrm{tol}\big)$ steps.

Proof. Each step keeps an interval on which h changes sign, so by the IVT every $[a_k, b_k]$ contains a root. (a_k) is increasing and bounded by b; (b_k) is decreasing and bounded by a; both converge (Thm 1.4b), and

since $b_k - a_k = (b-a)/2^k \to 0$ they have the same limit c. From $h(a_k)h(b_k) \le 0$ and continuity (Lemma 1.3), $h(c)^2 = \lim h(a_k)h(b_k) \le 0$, so $h(c) = 0$. Finally m_k and c both lie in $[a_k, b_k]$, and m_k is its midpoint. ∎

In our project, $[0, 2]$ with tolerance 10^{-12} needs $\log_2(2/10^{-12}) = 40.86$ — i.e. **41 steps**, exactly what the program prints.

1.6 Probability background

Normal distribution. $Z \sim N(0,1)$ has density $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and CDF $\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \varphi$. $X \sim N(m, s^2)$ means $X = m + sZ$. Φ is continuous and strictly increasing, and $\Phi(-z) = 1 - \Phi(z)$.

Quantile. For $q \in (0,1)$ the quantile $z_q = \Phi^{-1}(q)$ is the unique solution of $\Phi(z) = q$. Existence follows from the IVT, uniqueness from strict monotonicity. We need $z_{0.95} = 1.644854$.

Quantiles and monotone maps. If ψ is strictly increasing then the q-quantile of $\psi(X)$ is ψ(q-quantile of X).

Lemma 1.11 (lognormal moments). If $X \sim N(m, s^2)$ then $E[e^X] = e^{m + s^2/2}$.

Proof. $E[e^X] = \int \frac{1}{s\sqrt{2\pi}} e^{x - \frac{(x-m)^2}{2s^2}}\,dx$. Completing the square, $x - \frac{(x-m)^2}{2s^2} = m + \frac{s^2}{2} - \frac{(x-m-s^2)^2}{2s^2}$, and the remaining integral is that of the $N(m+s^2, s^2)$ density, which equals 1. ∎

Convergence in distribution. $X_n \Rightarrow X$ if $F_{X_n}(x) \to F_X(x)$ at every continuity point of F_X.

Central Limit Theorem (de Moivre–Laplace / Lindeberg form). Sums of many small independent random steps, centred and scaled, converge in distribution to a normal law. In particular if $J_n \sim \mathrm{Bin}(n, p_n)$ with $np_n(1-p_n) \to \infty$, then $\frac{J_n - np_n}{\sqrt{np_n(1-p_n)}} \Rightarrow N(0,1)$.

Kolmogorov–Smirnov distance. $D(F,G) = \sup_x |F(x) - G(x)|$. If the limit F is continuous, $F_n \to F$ pointwise implies $D(F_n, F) \to 0$ (Pólya's theorem).

Berry–Esseen theorem. For a normalised sum of n i.i.d. steps with finite third moment, $D(F_n, \Phi) \le C/\sqrt{n}$.

Law of large numbers. Averages of i.i.d. variables converge to the mean: $\frac{1}{n}\sum_{k=1}^{n} Y_k \to E[Y]$.

Part II — Module 1: Continuous compounding

2.1 Model

Invest P at nominal annual rate r, compounded n times per year. Each period the balance is multiplied by $1 + r/n$; after t years there are nt periods, so

$$A_n = P\left(1 + \frac{r}{n}\right)^{nt}. \tag{2}$$

Question: what happens as compounding becomes more and more frequent, $n \to \infty$?

2.2 The limit

> **Theorem 2.1.** $\lim_{n\to\infty} P\left(1 + \frac{r}{n}\right)^{nt} = Pe^{rt}$ for every $r > 0, t > 0$.

Proof. Write $A_n = P\exp(x_n)$ with $x_n = nt\ln(1 + r/n)$. Apply Lemma 1.5 with $u = r/n$:

$$nt \cdot \frac{r/n}{1 + r/n} \;<\; x_n \;<\; nt \cdot \frac{r}{n} \quad\Longleftrightarrow\quad \frac{rt}{1 + r/n} \;<\; x_n \;<\; rt.$$

The left side tends to rt, so by the squeeze theorem $x_n \to rt$. Since $\exp$ is continuous, Lemma 1.3 gives $A_n = Pe^{x_n} \to Pe^{rt}$. ∎

> **Proposition 2.2 (more frequent compounding always pays more, but boundedly).** A_n is strictly increasing in n and $A_n < Pe^{rt}$ for every n.

Proof. Consider $\phi(x) = x\ln(1 + r/x)$ for real $x > 0$. With $u = r/x$, $\phi'(x) = \ln(1 + u) - \frac{u}{1+u} > 0$ by the left inequality in Lemma 1.5. So ϕ is increasing and $A_n = Pe^{t\phi(n)}$ increases with n. The upper bound is the right inequality $x_n < rt$ from the previous proof. ∎

So (A_n) is increasing and bounded: it converges by monotone convergence, and Theorem 2.1 identifies the limit.

2.3 How fast? The rate of convergence

Using Lemma 1.6 with $u = r/n$:

$$x_n = nt\left(\frac{r}{n} - \frac{r^2}{2n^2} + O(n^{-3})\right) = rt - \frac{r^2 t}{2n} + O(n^{-2}).$$

Let $\varepsilon_n = rt - x_n = \frac{r^2 t}{2n} + O(n^{-2})$. Then

$$Pe^{rt} - A_n = Pe^{rt}\left(1 - e^{-\varepsilon_n}\right) = Pe^{rt}\left(\varepsilon_n + O(\varepsilon_n^2)\right) = \boxed{Pe^{rt}\,\frac{r^2 t}{2n} + O\!\left(\frac{1}{n^2}\right)}. \tag{3}$$

The gap shrinks like $1/n$: doubling the compounding frequency roughly halves the remaining gain.

2.4 Numbers ($P = 10{,}000$, $r = 10\%$, $t = 10$: $Pe^{rt} = 10{,}000\,e = 27{,}182.82$)

compounding	n	A_n	gap $Pe^{rt} - A_n$	estimate (3)
yearly	1	25,937.42	1,245.39	1,359.14
quarterly	4	26,850.64	332.18	339.79
monthly	12	27,070.41	112.40	113.26
weekly	52	27,156.73	26.09	26.14
daily	365	27,179.10	3.72	3.72
hourly	8,760	27,182.66	0.155	0.155
every minute	525,600	27,182.82	0.0026	0.0026

The estimate is poor for $n = 1$ (where $r/n = 0.1$ is not small, so the $O(n^{-2})$ term matters) and becomes essentially exact as n grows — itself an illustration of an asymptotic limit. On a log–log plot the gap is a straight line of slope -1.

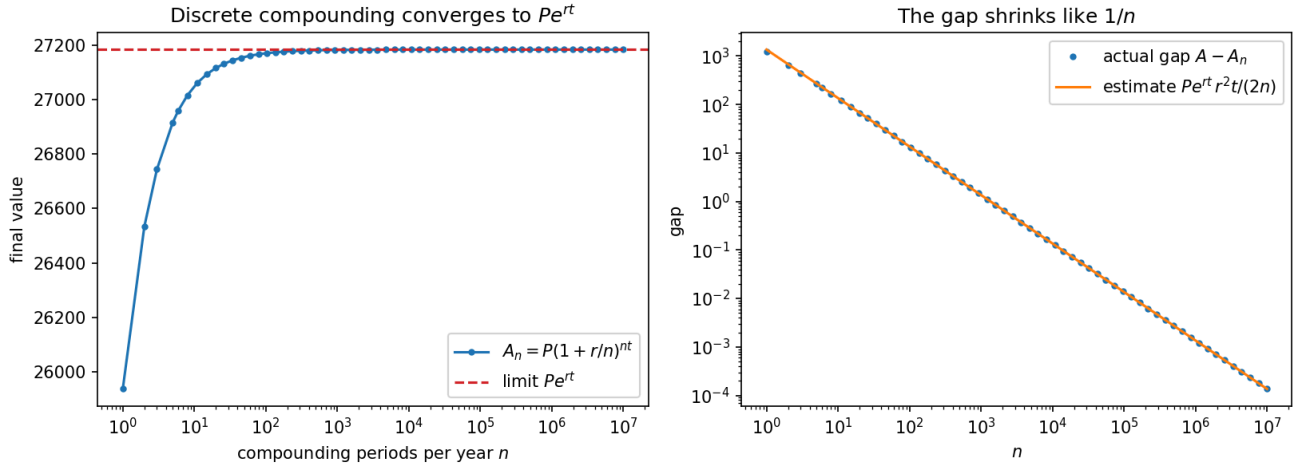


Figure 1. Left: A_n increases to the limit Pe^{rt}. Right: log–log plot of the gap, with the estimate $Pe^{rt}r^2t/(2n)$.

> **Real-life conclusion.** Switching from monthly to daily compounding gains 108.68; from daily to continuous only 3.72. Continuous compounding is the natural *idealisation* of frequent compounding, and its effective annual rate is $e^r - 1 = 10.517\%$ for $r = 10\%$. Modules 2–4 use e^{rt} as the growth of the risk-free asset.

Part III — Module 2: Binomial model → Black–Scholes

3.1 What is being priced

A **European call option** with strike K and maturity T gives the right (not the obligation) to buy the stock at price K at time T. Its value at maturity is the **payoff** $(S_T - K)^+ = \max(S_T - K, 0)$. A put pays $(K - S_T)^+$. Question: what is the fair price today?

3.2 One step: no-arbitrage pricing

In one step of length Δt, the stock moves from S to Su or Sd ($d < u$); money in the bank grows by $e^{r\Delta t}$. Let the option be worth V_u or V_d after the step. Build a portfolio of Δ shares and B in the bank that pays exactly the same in both states:

$$\Delta S u + B e^{r\Delta t} = V_u, \qquad \Delta S d + B e^{r\Delta t} = V_d.$$

Subtracting, $\Delta = \dfrac{V_u - V_d}{S(u-d)}$, and then $B = e^{-r\Delta t}\dfrac{uV_d - dV_u}{u-d}$. If the option were priced differently from the portfolio, one could buy the cheaper and sell the dearer for a riskless profit (an *arbitrage*). So

$$V = \Delta S + B = e^{-r\Delta t}\big[pV_u + (1-p)V_d\big], \qquad p = \frac{e^{r\Delta t} - d}{u - d}. \tag{4}$$

No arbitrage between stock and bank requires $d < e^{r\Delta t} < u$, i.e. $0 < p < 1$: p behaves like a probability, called the **risk-neutral probability**. Remarkably, the real-world drift μ does not appear.

3.3 N steps: the Cox–Ross–Rubinstein tree

Split $[0, T]$ into N steps, $\Delta t = T/N$, with $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$. After N steps with j up-moves the price is $Su^j d^{N-j} = Se^{\sigma\sqrt{\Delta t}(2j-N)}$. Applying (4) backwards through the tree (this is what the code does) is equivalent to the closed form

$$C_N = e^{-rT}\sum_{j=0}^{N}\binom{N}{j}p^j(1-p)^{N-j}\big(Su^j d^{N-j} - K\big)^+ = e^{-rT}\,E\big[(Se^{X_N} - K)^+\big],$$

where $X_N = \sigma\sqrt{\Delta t}\,(2J - N)$ and $J \sim \mathrm{Bin}(N, p)$ counts up-moves.

3.4 The limit $N \to \infty$

Step 1: expand p. With $\delta = \sqrt{\Delta t}$: $u = 1 + \sigma\delta + \tfrac{1}{2}\sigma^2\delta^2 + O(\delta^3)$, $d = 1 - \sigma\delta + \tfrac{1}{2}\sigma^2\delta^2 + O(\delta^3)$, $e^{r\Delta t} = 1 + r\delta^2 + O(\delta^4)$. Hence $u - d = 2\sigma\delta + O(\delta^3)$ and $e^{r\Delta t} - d = \sigma\delta + (r - \tfrac{1}{2}\sigma^2)\delta^2 + O(\delta^3)$, so

$$p = \frac{1}{2} + \frac{r - \sigma^2/2}{2\sigma}\sqrt{\Delta t} + O(\Delta t).$$

(For $N = 100$: $p = 0.50750$, and the formula gives $0.5 + 0.03/0.4 \cdot 0.1 = 0.5075$.)

Step 2: mean and variance of X_N. Since $E[J] = Np$, $\mathrm{Var}(J) = Np(1-p)$:

$$E[X_N] = \sigma\sqrt{\Delta t}\,N(2p-1) = \sigma\sqrt{\Delta t}\,N\frac{r - \sigma^2/2}{\sigma}\sqrt{\Delta t} + O(\sqrt{\Delta t}) = (r - \tfrac{1}{2}\sigma^2)T + O(\sqrt{\Delta t}),$$

$$\mathrm{Var}(X_N) = 4\sigma^2\Delta t\,Np(1-p) = \sigma^2 T \cdot 4p(1-p) \;\to\; \sigma^2 T \quad (\text{since } p \to \tfrac{1}{2}).$$

Step 3: CLT. X_N is a sum of N independent small steps $\pm\sigma\sqrt{\Delta t}$. By the CLT,

$$X_N \Rightarrow X \sim N\big((r-\tfrac12\sigma^2)T,\ \sigma^2 T\big).$$

Step 4: pass to the limit in the price. The payoff $x \mapsto (Se^x - K)^+$ is continuous, and its growth is controlled ($E[e^{2X_N}]$ stays bounded), which is enough to conclude $E[(Se^{X_N}-K)^+] \to E[(Se^X-K)^+]$. Therefore $\lim_{N\to\infty} C_N = e^{-rT}E[(Se^X-K)^+]$.

3.5 Computing the limit: the Black–Scholes formula

Let $m = (r-\frac12\sigma^2)T$, $s = \sigma\sqrt{T}$, $k = \ln(K/S)$. The option pays only when $X > k$, so

$$E[(Se^X-K)^+] = S\,E\big[e^X \mathbf{1}_{X>k}\big] - K\,P(X>k).$$

Second term: $P(X>k) = P\big(Z > \frac{k-m}{s}\big) = \Phi\big(\frac{m-k}{s}\big) = \Phi(d_2)$, with $d_2 = \dfrac{\ln(S/K) + (r-\frac12\sigma^2)T}{\sigma\sqrt{T}}$.

First term: completing the square as in Lemma 1.11, e^x times the $N(m,s^2)$ density equals $e^{m+s^2/2}$ times the $N(m+s^2,s^2)$ density. Hence $E[e^X\mathbf{1}_{X>k}] = e^{m+s^2/2}\,\Phi\big(\frac{m+s^2-k}{s}\big) = e^{rT}\Phi(d_1)$, with $d_1 = d_2 + \sigma\sqrt{T}$.

Multiplying by e^{-rT}:

$$C_{BS} = S\,\Phi(d_1) - Ke^{-rT}\Phi(d_2), \qquad d_{1,2} = \frac{\ln(S/K) + (r \pm \frac12\sigma^2)T}{\sigma\sqrt{T}}.$$

For the put, use **put–call parity**: since $(x-K)^+ - (K-x)^+ = x-K$, we get $C - P = S - Ke^{-rT}$, so $P_{BS} = Ke^{-rT}\Phi(-d_2) - S\Phi(-d_1)$.

3.6 Numbers and rate of convergence

With $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$: $d_1 = 0.35$, $d_2 = 0.15$, $C_{BS} = 10.450584$.

N	1	2	5	10	50	100	500	1000
C_N	12.1623	9.5405	10.8059	10.2534	10.4107	10.4306	10.4466	10.4486
$\lvert C_N - C_{BS}\rvert$	1.7117	0.9101	0.3554	0.1972	0.0399	0.0200	0.0040	0.0020

The error behaves like c/N: the program gives $N\cdot|C_N - C_{BS}| \approx 2.00$ for $N = 100, 1000$ but ≈ 1.75 for $N = 101, 1001$. The constant depends on the parity of N because the position of the strike K relative to the tree's final nodes changes with N. This is the **even/odd oscillation** visible in Figure 2.

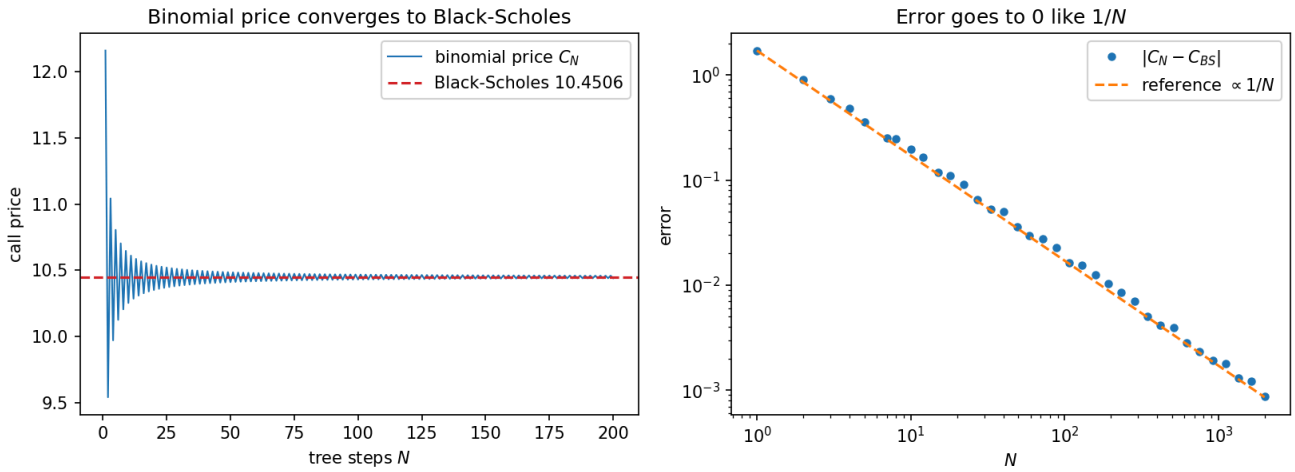


Figure 2. Left: binomial price vs. number of steps, oscillating towards Black–Scholes. Right: the error decays like $1/N$.

Part IV — Module 3: Random walk → Geometric Brownian Motion

4.1 The continuous model: Geometric Brownian Motion

The standard model of a stock price is

$$S_t = S_0 \exp\Big(\big(\mu - \tfrac{1}{2}\sigma^2\big)t + \sigma B_t\Big),$$

where B_t is Brownian motion ($B_t \sim N(0,t)$, independent increments). So the log-return is normal:

$$\ln \frac{S_T}{S_0} \sim N\Big(\big(\mu - \tfrac{1}{2}\sigma^2\big)T,\ \sigma^2 T\Big). \tag{5}$$

Why $-\frac{1}{2}\sigma^2$? By Lemma 1.11, $E[S_T] = S_0 e^{(\mu-\sigma^2/2)T + \sigma^2 T/2} = S_0 e^{\mu T}$. So μ is the growth rate of the *average* price, while the typical (median) price grows only at $\mu - \frac{1}{2}\sigma^2$. This "volatility drag" returns in Module 4 as the $-\frac{1}{2}f^2\sigma^2$ term.

4.2 The discrete model: a biased coin-flip walk

Take n steps of length $\Delta t = T/n$. At each step the log-price moves up or down by $\sigma\sqrt{\Delta t}$, up with probability

$$p = \frac{1}{2} + \frac{1}{2}\cdot\frac{\mu - \sigma^2/2}{\sigma}\sqrt{\Delta t}.$$

(We need $0 \le p \le 1$, i.e. $\Delta t \le \sigma^2/(\mu - \sigma^2/2)^2 = 6.25$ years — always true here.) After n steps, with $J \sim \mathrm{Bin}(n,p)$ up-moves, $\ln(S_T/S_0) = \sigma\sqrt{\Delta t}\,(2J - n)$.

> **Proposition 4.1.** For every n:
>
> $$E\Big[\ln\frac{S_T}{S_0}\Big] = \big(\mu - \tfrac{1}{2}\sigma^2\big)T \ \text{(exactly)}, \qquad \mathrm{Var}\Big[\ln\frac{S_T}{S_0}\Big] = \sigma^2 T\Big(1 - \frac{(\mu-\sigma^2/2)^2}{\sigma^2}\Delta t\Big) \xrightarrow[\Delta t\to 0]{} \sigma^2 T.$$

Proof. Mean: $\sigma\sqrt{\Delta t}\,n(2p-1) = \sigma\sqrt{\Delta t}\,n\,\frac{\mu-\sigma^2/2}{\sigma}\sqrt{\Delta t} = (\mu - \frac{1}{2}\sigma^2)n\Delta t$. Variance: $4\sigma^2\Delta t\,np(1-p) = \sigma^2 T\cdot 4p(1-p)$ and $4p(1-p) = 1 - (2p-1)^2$. ∎

Example $n = 12$: variance 0.039467 vs. limit 0.04. By the CLT the whole distribution converges: $\ln(S_T/S_0) \Rightarrow$ the GBM law (5). **The random walk becomes GBM in the limit $\Delta t \to 0$.**

4.3 Measuring the convergence exactly

We measure the distance between the walk's CDF F_n and the GBM CDF F by the Kolmogorov–Smirnov distance $D_n = \sup_x |F_n(x) - F(x)|$. Because F_n is a step function with jumps at the lattice points $x_k = \sigma\sqrt{\Delta t}(2k - n)$, and F is increasing, the supremum is attained at a jump, either just before it or at it:

$$D_n = \max_{0\le k\le n} \max\big(|F_n(x_k) - F(x_k)|,\ |F_n(x_k^-) - F(x_k)|\big).$$

This is computed from the exact binomial probabilities, so there is **no simulation noise**. Binomial coefficients like $\binom{25200}{12600} \approx 10^{7583}$ overflow a computer, so they are built in logarithms using $\ln\binom{n}{k} = \ln\binom{n}{k-1} + \ln\frac{n-k+1}{k}$.

4.4 Why exactly $1/\sqrt{n}$?

Upper bound: Berry–Esseen gives $D_n \le C/\sqrt{n}$. **Lower bound:** at a lattice point F_n jumps by $P(J=k)$, but the continuous F barely moves there, so F is at least half a jump away from F_n on one side: $D_n \gtrsim \frac{1}{2}\max_k P(J=k)$. By the local CLT (Stirling's formula), $\max_k P(J=k) \approx 1/\sqrt{2\pi np(1-p)}$. Therefore

$$D_n \approx \frac{1}{2\sqrt{2\pi np(1-p)}} \approx \frac{0.399}{\sqrt{n}}.$$

n (steps per year)	1	12	252	25,200
exact D_n	0.4257	0.1192	0.02530	0.002515
$1/(2\sqrt{2\pi np(1-p)})$	0.4353	0.1159	0.02514	0.002513

The prediction matches the computation almost perfectly, and explains the slope $-\frac{1}{2}$ on the log–log plot in Figure 3.

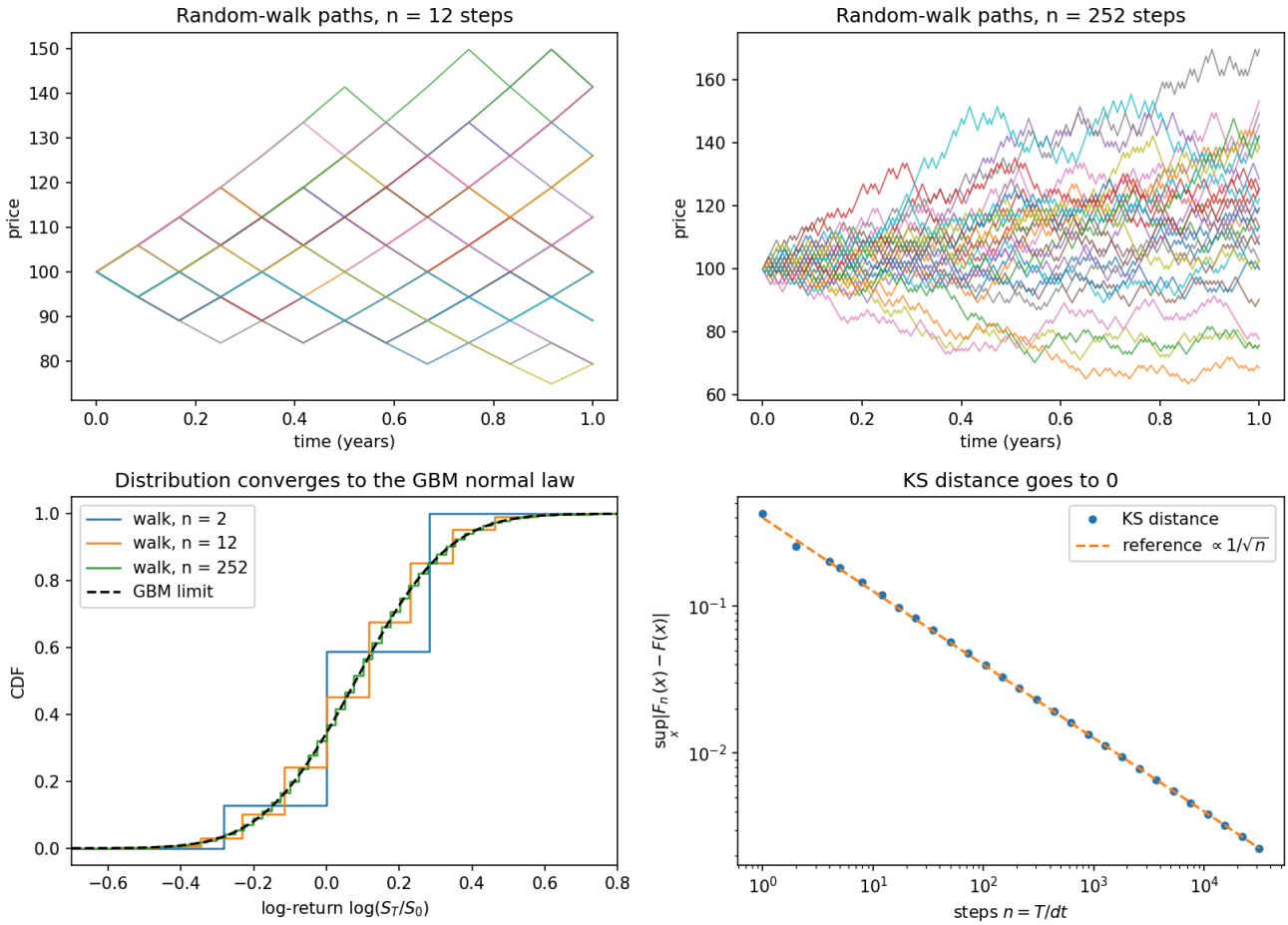


Figure 3. Top: walk paths with 12 and 252 steps. Bottom left: walk CDFs (steps) approaching the GBM CDF (dashed). Bottom right: $D_n \propto 1/\sqrt{n}$.

Part V — Module 4: Risk optimization

5.1 The portfolio and its limit

Hold a fraction f of wealth in the stock and $1-f$ in the bank, and **rebalance** back to the fraction f every Δt. ($f>1$ means borrowing at rate r to buy extra stock; $f<0$ would be short-selling.) Over one step the stock's log-return is $Y_k=(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}\,Z_k$ with $Z_k\sim N(0,1)$ i.i.d. (GBM, Module 3), and the bank returns $\rho=e^{r\Delta t}-1$ (Module 1). Wealth evolves as

$$\frac{W_{k+1}}{W_k}=1+x_k,\qquad x_k=f\,(e^{Y_k}-1)+(1-f)\rho.$$

Theorem 5.1 (continuous-rebalancing limit). As $\Delta t\to 0$,

$$\ln\frac{W_T}{W_0}\;\Rightarrow\; g(f)\,T+f\sigma\sqrt{T}\,Z,\qquad Z\sim N(0,1),\qquad \boxed{g(f)=r+f(\mu-r)-\tfrac{1}{2}f^2\sigma^2}.$$

Proof. All expansions are in powers of $\sqrt{\Delta t}$; $Y_k=O(\sqrt{\Delta t})$. By Lemma 1.6, $e^{Y_k}-1=Y_k+\frac{1}{2}Y_k^2+O(\Delta t^{3/2})=(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}Z_k+\frac{1}{2}\sigma^2Z_k^2\Delta t+O(\Delta t^{3/2})$, and $\rho=r\Delta t+O(\Delta t^2)$. Then $x_k^2=f^2\sigma^2Z_k^2\Delta t+O(\Delta t^{3/2})$ and $\ln(1+x_k)=x_k-\frac{1}{2}x_k^2+O(\Delta t^{3/2})$, so

$$\ln(1+x_k)=r\Delta t+f(\mu-r)\Delta t+f\sigma\sqrt{\Delta t}Z_k\underbrace{-\tfrac{1}{2}f\sigma^2\Delta t+\tfrac{1}{2}f\sigma^2Z_k^2\Delta t}_{\text{from the stock}}\underbrace{-\tfrac{1}{2}f^2\sigma^2Z_k^2\Delta t}_{\text{from }\ln}+O(\Delta t^{3/2}).$$

Sum over the $n=T/\Delta t$ steps: (i) $\sum Z_k^2\Delta t=T\cdot\frac{1}{n}\sum Z_k^2\to T$ by the law of large numbers; (ii) $\sum f\sigma\sqrt{\Delta t}Z_k\sim f\sigma\sqrt{T}\,Z$ exactly (a sum of independent normals); (iii) $n\cdot O(\Delta t^{3/2})=O(\sqrt{\Delta t})\to 0$. The two "stock" terms cancel in the limit, leaving $\ln(W_T/W_0)\to\big(r+f(\mu-r)-\frac{1}{2}f^2\sigma^2\big)T+f\sigma\sqrt{T}Z$. ∎

This is Itô's lemma in elementary form: the $Z_k^2\Delta t$ terms do not vanish (their sum tends to T), which is why a σ^2 correction appears. The $-\frac{1}{2}f^2\sigma^2$ term is the volatility drag of the whole portfolio.

5.2 Why $g(f)$ is "the growth rate"

From Theorem 5.1, $W_T=W_0\exp\big(g(f)T+f\sigma\sqrt{T}Z\big)$, so

$$\frac{1}{T}\ln\frac{W_T}{W_0}=g(f)+\frac{f\sigma Z}{\sqrt{T}}\;\xrightarrow[T\to\infty]{}\;g(f).$$

Over a long horizon wealth grows like $e^{g(f)T}$, whatever the luck. Also $g(f)T=E[\ln(W_T/W_0)]$, the expected log-wealth.

5.3 Maximum growth: the Kelly fraction

g is a quadratic polynomial, so it is continuous and differentiable:

$$g'(f)=\lim_{h\to 0}\frac{g(f+h)-g(f)}{h}=(\mu-r)-f\sigma^2,\qquad g''(f)=-\sigma^2<0.$$

g is strictly concave, so the unique maximum is where $g'(f)=0$:

$$f^* = \frac{\mu - r}{\sigma^2} = \frac{0.07}{0.04} = 1.75, \qquad g(f^*) = r + \frac{(\mu - r)^2}{2\sigma^2} = 9.125\%.$$

This is the **Kelly (Merton) fraction**. Note $g(f) - r = f\big((\mu - r) - \tfrac{1}{2} f\sigma^2\big)$ vanishes at $f = 0$ and $f = 2f^* = 3.5$: beyond $3.5\times$ leverage you grow *slower than the bank*, despite taking huge risk.

The derivative as a limit, numerically

Because g is quadratic, the difference quotient can be computed exactly: $\frac{g(f+h)-g(f)}{h} = g'(f) - \frac{\sigma^2}{2}h$. At $f = 1$ (where $g'(1) = 0.03$):

h	0.1	0.01	0.001	10^{-4}	10^{-5}	10^{-6}
quotient	0.028	0.0298	0.02998	0.029998	0.0299998	0.02999998

The error $-0.02h$ goes to 0 linearly in h. The program also finds f^* without the formula: g' is continuous with $g'(0) = 0.07 > 0$ and $g'(10) = -0.33 < 0$, so by the IVT bisection on g' finds the root 1.750000. (It uses the central quotient $\frac{g(f+h)-g(f-h)}{2h}$, which is *exact* for quadratics.)

The problem with Kelly: $f^* = 1.75$ means borrowing 75% of your wealth, and (see below) its one-year 95% VaR is 38.4%.

5.4 Measuring risk: Value-at-Risk

Definition 5.2. The one-year loss (as a fraction of initial wealth) is $\mathcal{L} = 1 - W_T/W_0$. The **Value-at-Risk** at level α is the loss that is exceeded only with probability $1 - \alpha$: $P(\mathcal{L} > \mathrm{VaR}_\alpha) = 1 - \alpha$.

Proposition 5.3. For $f \geq 0$,

$$\mathrm{VaR}_\alpha(f) = 1 - \exp\big(g(f)T - z_\alpha f\sigma\sqrt{T}\big), \qquad z_\alpha = \Phi^{-1}(\alpha).$$

Proof. VaR is the α-quantile of $\mathcal{L}$, i.e. $1-$(the $(1-\alpha)$-quantile of W_T/W_0). $W_T/W_0 = \exp(Y)$, $Y \sim N(g(f)T, f^2\sigma^2 T)$, and $\exp$ is increasing, so its $(1-\alpha)$-quantile is $\exp(g(f)T + z_{1-\alpha} f\sigma\sqrt{T})$ with $z_{1-\alpha} = -z_\alpha$. ∎

$z_{0.95} = 1.644854$ is found by bisection on $\Phi(z) - 0.95$ over $[-10, 10]$ — another IVT application. Note $\mathrm{VaR}(0) = 1 - e^{rT} = -3.05\%$: the all-bank portfolio has a negative VaR (a guaranteed gain).

5.5 The risk constraint: continuity and the IVT

The rule is $\mathrm{VaR}(f) \leq L = 0.15$. Define

$$h(f) = \mathrm{VaR}(f) - L.$$

- **Continuity:** h is a composition of a polynomial and $\exp$, so it is continuous on $[0, 2]$.
- **Signs:** $h(0) = (1 - e^{0.03}) - 0.15 = -0.1805 < 0$ and $h(2) = 0.4333 - 0.15 = 0.2833 > 0$.

By the Intermediate Value Theorem there exists $f_{VaR} \in (0, 2)$ with $h(f_{VaR}) = 0$: an allocation whose risk is *exactly* at the limit.

Proposition 5.4 (uniqueness and the feasible set). If $h(0) < 0$, then for **every** horizon T the root f_{VaR} is the only root on $[0, \infty)$, and $\{f \geq 0 : \mathrm{VaR}(f) \leq L\} = [0, f_{VaR}]$. If moreover $(\mu - r)\sqrt{T} < z_\alpha \sigma$, VaR is strictly increasing on $[0, \infty)$.

Proof. $\mathrm{VaR}(f) = 1 - e^{\eta(f)}$ with $\eta(f) = rT + f\big((\mu - r)T - z_\alpha \sigma \sqrt{T}\big) - \frac{1}{2} f^2 \sigma^2 T$, so $\mathrm{VaR}(f) \leq L \iff \eta(f) \geq \ln(1 - L)$, and $\eta(f) = \ln(1 - L)$ is the quadratic equation $\frac{1}{2}\sigma^2 T\, f^2 + \big(z_\alpha \sigma \sqrt{T} - (\mu - r)T\big) f + \big(\ln(1 - L) - rT\big) = 0$. Its constant term is negative, because $h(0) < 0 \iff rT > \ln(1 - L)$. The product of the two roots is the constant term divided by $\frac{1}{2}\sigma^2 T > 0$, so it is negative: one root is negative and exactly one is positive. Between the roots the parabola is below 0, that is $\eta > \ln(1 - L)$; so for $f \geq 0$ the rule holds exactly on $[0, f_{VaR}]$. For the monotonicity, $\eta'(f) = (\mu - r)T - z_\alpha \sigma \sqrt{T} - f\sigma^2 T < 0$ for $f \geq 0$ under the extra assumption, so VaR increases. Here $(\mu - r)\sqrt{T} = 0.07 < z\sigma = 0.329$. For horizons beyond $(z\sigma/(\mu - r))^2 \approx 22$ years VaR first *falls* and then rises, but the root is still unique. ∎

Bisection on h over $[0, 2]$

k	a_k	b_k	midpoint	h(mid)	keep
1	0	2	1	+0.0704	left half
2	0	1	0.5	−0.0508	right half
3	0.5	1	0.75	+0.0109	left half
4	0.5	0.75	0.625	−0.0196	right half
5	0.625	0.75	0.6875	−0.0043	right half
6	0.6875	0.75	0.71875	+0.0034	left half
7	0.6875	0.71875	0.703125	−0.0005	right half
8	0.703125	0.71875	0.710938	+0.0015	left half
⋮					
41	0.705014	0.705014	**0.705014**	10^{-13}	stop

Cross-check with a closed form

In this model $h(f) = 0$ is a quadratic equation: $\eta(f) = \ln(1 - L)$, i.e. $\frac{1}{2}\sigma^2 T f^2 + \big(z_\alpha \sigma \sqrt{T} - (\mu - r)T\big) f + \big(\ln(1 - L) - rT\big) = 0$, i.e. $0.02 f^2 + 0.258971 f - 0.192519 = 0$, whose positive root is $f = 0.7050142$ — identical to bisection. **Why use bisection then?** Because it only needs continuity. With realistic risk models (historical returns, fat-tailed distributions, Monte Carlo VaR) there is no formula, but the IVT argument and bisection still work unchanged.

5.6 The optimal allocation

Problem: maximise $g(f)$ subject to $0 \le f \le f_{max}$ and $\mathrm{VaR}(f) \le L$. By Proposition 5.4 the feasible set is the closed interval $[0, c]$ with $c = \min(f_{VaR}, f_{max})$. Since g is continuous, the Extreme Value Theorem guarantees a maximiser exists. Since g is concave with peak at f^*, the maximiser on $[0, c]$ is f^* if $f^* \le c$, and otherwise c (because $g' > 0$ on $[0, f^*)$, g is increasing there):

$$f_{opt} = \min\left(f^*,\ f_{VaR},\ f_{max}\right) = \min(1.75,\ 0.705,\ 2) = 0.705.$$

Invest 70.5% in the stock and 29.5% in the bank. Growth $g(0.705) = 6.94\%$ per year (vs. 9.13% for Kelly) with VaR exactly 15% (vs. 38.4% for Kelly). The risk constraint is *binding*.

Sensitivity: how the answer depends on the inputs

loss limit L	5%	10%	15%	20%	25%	40%
f_{opt}	0.307	0.503	**0.705**	0.913	1.128	1.750
growth $g(f_{opt})$	4.96%	6.02%	6.94%	7.72%	8.35%	9.13%

More risk tolerance allows more stock and more growth. At $L = 40\%$, $f_{VaR} = 1.83 > f^*$: the constraint is no longer binding and the answer becomes Kelly — the "min" switches branch. With $L = 15\%$, higher volatility lowers the allocation: $\sigma = 15\% \Rightarrow f_{opt} = 1.023$; $\sigma = 25\% \Rightarrow 0.538$; $\sigma = 30\% \Rightarrow 0.435$.

5.7 Verification by Monte Carlo simulation

The program simulates 20,000 one-year wealth paths with discrete rebalancing (the model of §5.1, before the limit) and computes the empirical 5% quantile of the loss. By the law of large numbers (more precisely Glivenko–Cantelli), the empirical quantile converges to the true one, with standard error

$$\mathrm{SE} \approx \frac{\sqrt{\alpha(1-\alpha)/N}}{\text{density of } W_T/W_0 \text{ at the quantile}} = \frac{\sqrt{0.05 \cdot 0.95/20000}}{0.861} \approx 0.18\%.$$

rebalancing	yearly	quarterly	monthly	weekly	daily
simulated VaR	14.63%	15.06%	14.76%	14.89%	14.91%

All values are within about two standard errors of the analytical 15%; yearly rebalancing additionally differs from the continuous model because $\Delta t = 1$ is far from the limit. The simulation confirms Theorem 5.1 and Proposition 5.3.

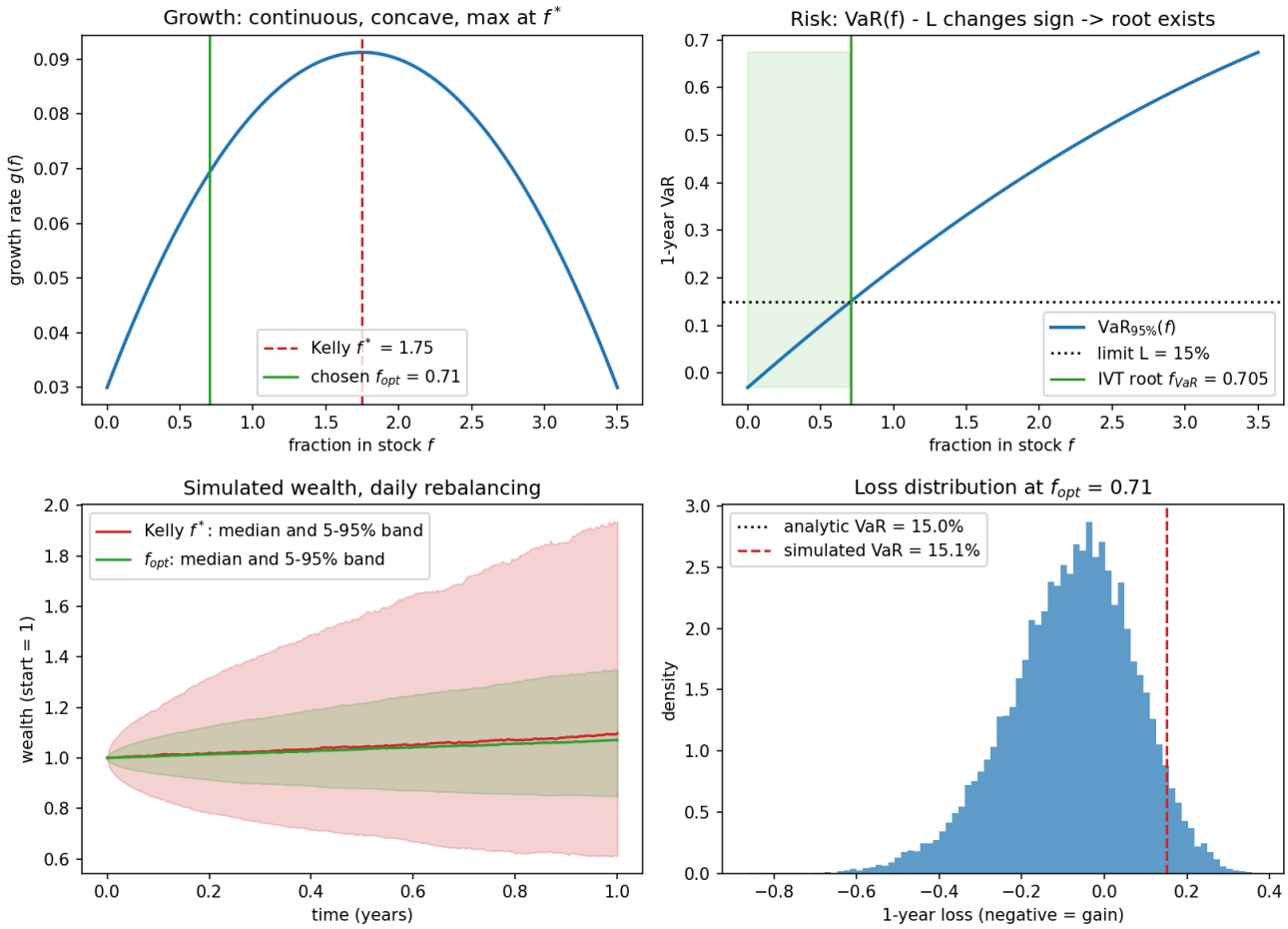


Figure 4. Top left: concave growth $g(f)$, max at Kelly f^*. Top right: VaR(f) crosses the limit L at the IVT root. Bottom left: simulated wealth bands — Kelly is much wider. Bottom right: simulated loss distribution at f_{opt}.

5.8 How bad are the bad years? Expected Shortfall

VaR says where the worst $1-\alpha$ of years *start*, not how bad they are. The **Expected Shortfall** is their average loss.

Proposition 5.5 (Expected Shortfall). With $Y = \ln(W_T/W_0) \sim N(m, s^2)$, $m = g(f)T$, $s = f\sigma\sqrt{T}$ and the VaR point $q = m - z_\alpha s$,

$$\mathrm{ES}_\alpha(f) = 1 - \frac{E\left[e^Y \mathbf{1}_{\{Y \le q\}}\right]}{1-\alpha} = 1 - e^{m+s^2/2}\,\frac{\Phi(-z_\alpha - s)}{1-\alpha} \;\ge\; \mathrm{VaR}_\alpha(f).$$

Proof. As in Lemma 1.11, e^y times the $N(m, s^2)$ density equals $e^{m+s^2/2}$ times the $N(m+s^2, s^2)$ density, so $E[e^Y \mathbf{1}_{\{Y\le q\}}] = e^{m+s^2/2}\,\Phi\left(\frac{q-m-s^2}{s}\right) = e^{m+s^2/2}\,\Phi(-z_\alpha - s)$. On the event $\{Y \le q\}$ the loss is at least the VaR, so its average is too. ∎

At $f_{VaR} = 0.705$: VaR = 15%, but **ES = 19.76%** (a simulation with daily rebalancing gives 19.91%). Since $m + s^2/2 = (r + f(\mu - r))T$ is linear in f and $\ln\Phi$ is concave (Φ is log-concave), $\ln(1 - \mathrm{ES}(f))$ is concave; so, exactly as in Proposition 5.4, the rule $\mathrm{ES}(f) \le L$ holds on an interval $[0, f_{ES}]$ with one boundary point. Bisection gives $f_{ES} = 0.547$: the stricter rule "lose at most 15% *on average* in the worst 5% of years".

Part VI — Module 4b: Fat tails with rare crashes

6.1 Model

The bell curve makes a really bad year almost impossible. Now crashes arrive at rate $\lambda = 0.1$ per year (about once in 10 years) and multiply the stock by $1 + J$ with $J = -25\%$. Between crashes the stock is a GBM with drift $\mu - \lambda J$, so the average return is unchanged: $E[dS/S] = (\mu - \lambda J)\,dt + J \cdot \lambda\,dt = \mu\,dt$. With a fraction f in the stock, rebalanced continuously, a crash multiplies wealth by $1 + fJ$, so with $N \sim \text{Poisson}(\lambda T)$ crashes

$$\ln \frac{W_T}{W_0} = \big(g(f) - \lambda J f\big)T + N \ln(1 + fJ) + f\sigma\sqrt{T}\,Z.$$

6.2 A new limit: coin-flips become the Poisson law

> **Theorem 6.1 (Poisson limit).** If a crash happens in each of n steps of a year with probability λ/n, the number of crashes $B_n \sim \text{Bin}(n, \lambda/n)$ satisfies $\lim_{n\to\infty} P(B_n = k) = e^{-\lambda}\dfrac{\lambda^k}{k!}$ for every k.

Proof. $P(B_n = k) = \dfrac{n(n-1)\cdots(n-k+1)}{n^k} \cdot \dfrac{\lambda^k}{k!} \cdot \Big(1 - \dfrac{\lambda}{n}\Big)^n \cdot \Big(1 - \dfrac{\lambda}{n}\Big)^{-k}$. The first factor is a product of k terms that each tend to 1, and so does the last one. The third tends to $e^{-\lambda}$: this is Theorem 2.1 with $r = -\lambda$, whose proof works unchanged for a negative rate because Lemma 1.5 also holds for $-1 < u < 0$. ∎

The largest difference between the two probability tables is $9.5 \cdot 10^{-3}$ for $n = 1$, $3.4 \cdot 10^{-5}$ for $n = 252$ and $8.6 \cdot 10^{-7}$ for $n = 10{,}000$.

6.3 The VaR of a mixture: the IVT twice

Given the number of crashes $N = k$, log-wealth is normal, so its CDF is a **mixture of bell curves**:

$$F(x) = \sum_{k \ge 0} P(N = k)\,\Phi\left(\frac{x - (g(f) - \lambda J f)T - k\ln(1 + fJ)}{f\sigma\sqrt{T}}\right).$$

F is continuous and strictly increasing from 0 to 1, so $F(x) = 1 - \alpha$ has exactly one solution (IVT + monotonicity), which bisection finds; then $\text{VaR}_{crash}(f) = 1 - e^x$. It has no formula, but it is continuous in f, with $h(0) = -0.180 < 0$ and $h(2) = 0.361 > 0$ for $h(f) = \text{VaR}_{crash}(f) - L$. So a second bisection, whose function is itself a bisection, finds $f_{crash} = 0.617447$ (VaR_{crash} is increasing here, checked on a fine grid, so this is the only root). The sum is cut at $k = 30$; the Poisson weights after that are far below rounding error.

6.4 Same average return, lower growth

> **Proposition 6.2.** The long-run growth with crashes is $g_{crash}(f) = E[\ln(W_T/W_0)]/T = g(f) - \lambda J f + \lambda \ln(1 + fJ)$, and $g_{crash}(f) < g(f)$ for every $f \neq 0$.

Proof. $E[N] = \lambda T$ gives the formula. By Lemma 1.5, $\ln(1+x) < x$ for $x \neq 0$; with $x = fJ$ this gives $\lambda \ln(1+fJ) < \lambda fJ$. ∎

A big loss hurts more than the extra drift of the good years helps: at $f = 0.705$ the growth falls from 6.94% to 6.76% a year, although the average return is the same. g_{crash} is strictly concave ($g''_{crash} = -\sigma^2 - \lambda J^2/(1+fJ)^2 < 0$), and its slope $(\mu - r) - f\sigma^2 - \lambda J + \lambda J/(1+fJ)$ is continuous, equals $\mu - r > 0$ at $f = 0$ and tends to $-\infty$ as $1 + fJ \to 0$. By the IVT bisection finds the **crash-aware Kelly fraction** $f^*_{crash} = 1.4098$ (instead of 1.75). The decision is again $f_{opt} = \min(f^*_{crash}, f_{crash}, f_{max}) = 0.617$.

6.5 Results

at $f = 0.705$	bell curve only	with rare crashes
95% VaR	15.00%	17.47%
99% VaR	22.79%	27.29%
average loss in the worst 5% (ES)	19.76%	23.48%
years losing more than 15%	1 in 20	1 in 14
typical growth per year	6.94%	6.76%

The ES with crashes uses Proposition 5.5 for each bell curve of the mixture below the VaR point. A simulation with a crash coin-flip in every step gives a VaR at f_{crash} of 14.46% (yearly steps), 14.97% (monthly), 14.96% (weekly) and 15.04% (daily): the binomial → Poisson limit at work.

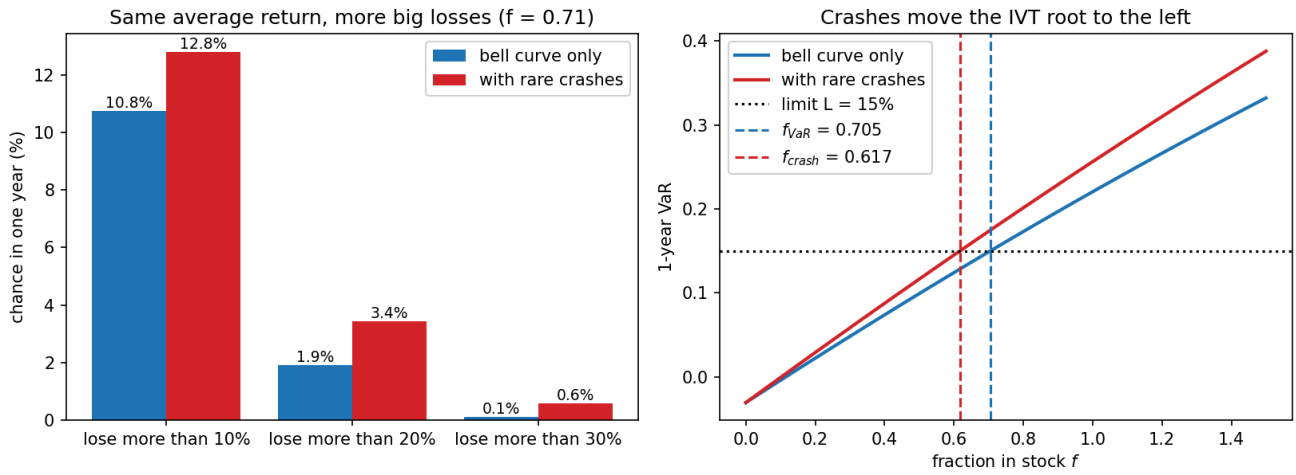


Figure 5. Left: the chance of big losses at $f = 0.705$, with and without crashes. Right: crashes move the IVT root to the left.

Part VII — Module 5: The best mix of stocks, gold and bonds

7.1 Model

Weights $w = (w_1, w_2, w_3)$ go to stocks, gold and bonds, and $1 - \sum w_i$ to the bank, with no borrowing ($\sum w_i \le 1$) and no short selling ($w_i \ge 0$). With the covariance matrix $C_{ij} = \rho_{ij}\sigma_i\sigma_j$ and continuous rebalancing, exactly as in Theorem 5.1, log-wealth is normal with

$$g(w) = r + w \cdot (\mu - r) - \tfrac{1}{2} w^\top C w, \qquad \mathrm{sd}(w) = \sqrt{w^\top C w}, \qquad \mathrm{VaR}(w) = 1 - e^{\,g(w)T - z_\alpha \,\mathrm{sd}(w)\sqrt{T}}.$$

> **Lemma 7.1 (one mix is one fund).** For a mix d of the risky assets ($d_i \ge 0$, $\sum d_i = 1$) and $w = fd$, $g(w) = r + f(\mu_d - r) - \tfrac{1}{2} f^2 \sigma_d^2$ and $\mathrm{sd}(w) = f\sigma_d$, with $\mu_d = r + d \cdot (\mu - r)$ and $\sigma_d = \sqrt{d^\top C d}$.

Proof. $w \cdot (\mu - r) = f\, d \cdot (\mu - r) = f(\mu_d - r)$ and $w^\top C w = f^2\, d^\top C d = f^2 \sigma_d^2$. ∎

So each fund is Module 4 with one "stock" (μ_d, σ_d), and its best share is $f(d) = \min(f_d^*, f_{VaR}(d), 1)$, with $f_{VaR}(d)$ from the IVT + bisection. Every allowed w is $f\,d$ for some fund d and $f \in [0, 1]$, so the best mix is the best fund.

7.2 Exactly one best mix

> **Theorem 7.2 (convexity).** The allowed mixes with $\mathrm{VaR}(w) \le L$ form a convex set, and the growth g has exactly one maximum on it.

Proof. g is strictly concave because its Hessian $-C$ is negative definite (the correlation matrix has leading minors $1, 1, 0.87 > 0$). $\mathrm{sd}(w) = \|C^{1/2} w\|$ is a norm, hence convex (triangle inequality), so $q(w) = g(w)T - z_\alpha \,\mathrm{sd}(w)\sqrt{T}$ is concave for $\alpha > \tfrac{1}{2}$. $\mathrm{VaR}(w) \le L \iff q(w) \ge \ln(1 - L)$ is a superlevel set of a concave function, so it is convex, and so is its intersection with $\{w \ge 0,\ \sum w_i \le 1\}$. This set is closed and bounded, so by the Extreme Value Theorem g has a maximum on it; if there were two, g at their midpoint would be larger (strict concavity), a contradiction. ∎

7.3 Searching on a grid, and the limit as the step goes to 0

Trying every fund on the grid with steps of 1% (5,151 funds, so 5,151 bisections) gives **73% stocks, 4% gold, 23% bonds** with growth 7.46824% and VaR 14.9953%: 0.53% a year more than stocks + bank (6.94%) for the same risk. But a grid only finds the best *grid* fund.

> **Theorem 7.3 (grid limit).** Let $G(d) = g(f(d)\,d)$ be the growth of fund d and G_s its maximum over the grid with step $1/s$. If s divides s', then $G_s \le G_{s'}$; and $G_s \to \max G$ as $s \to \infty$.

Proof. If $s \mid s'$, every point of the coarse grid is also a point of the fine one, so the fine maximum cannot be smaller. G is continuous on the triangle: f_d^* and $f_{VaR}(d)$ (the positive root of a quadratic whose coefficients depend continuously on d, Proposition 5.4) are continuous, and so is their minimum with 1.

The triangle is closed and bounded, so G is uniformly continuous; every point lies within $1/s$ of a grid point, so $\max G - G_s \to 0$. ∎

So the doubling sequence $G_{25} \le G_{50} \le G_{100} \le \dots$ increases to the exact maximum. For the bell curve $f_{VaR}(d)$ is the root of the quadratic of Proposition 5.4, so even grids with a million funds take a second:

grid step	mixes	best mix (stocks / gold / bonds)	growth g
4%	351	72 / 12 / 16	7.45461%
2%	1,326	72 / 12 / 16	7.45461%
1%	5,151	73 / 4 / 23	7.46824%
0.5%	20,301	72.98 / 4.50 / 22.49	7.46828%
0.25%	80,601	73 / 4.25 / 22.75	7.46875%
0.125%	321,201	73 / 4.25 / 22.75	7.46875%
0.0625%	1,282,401	72.94 / 5 / 22.06	7.46882%
$\to 0$	(zoom in)	72.9 / 4.9 / 22.2	**7.46887%**

The last row zooms in on the best fund with a small grid whose step halves 40 times. At the exact best both rules bind: all the money is invested and the VaR is exactly 15%.

7.4 A flat top

The growth hardly changes along the 15% boundary: every fund on the 1% grid within 0.01% a year of the best has 73% stocks, but anywhere from 1% to 7% gold (bonds 20–26%). So the precise gold share is not a meaningful answer; **about 73% stocks**, the rest mostly bonds, is. Bonds replace the bank because they pay a bit more and move a little against stocks. A simulation with daily rebalancing gives a VaR of 14.95% for the best mix.

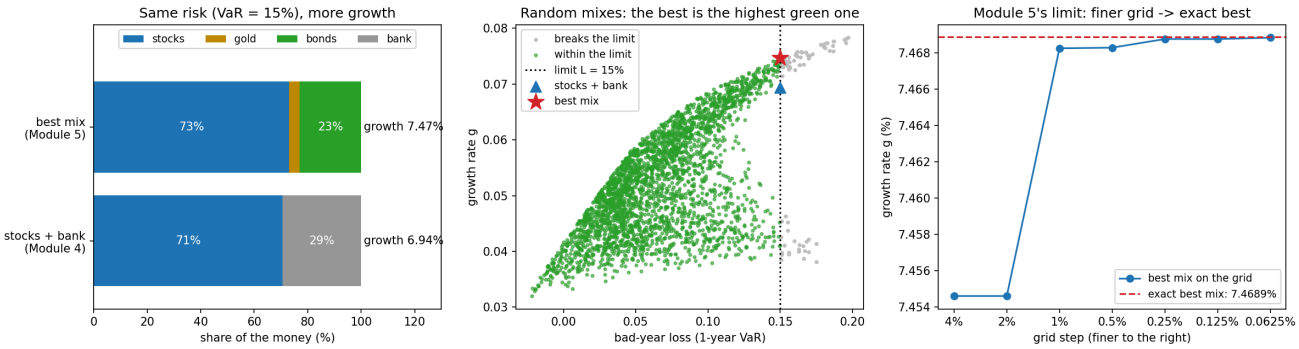


Figure 6. Left: same risk, more growth than stocks + bank. Middle: random mixes; the best is the highest one within the limit. Right: finer grids find better mixes and converge to the exact best.

Part VIII — From mathematics to code

Mathematics	Python (`src/`)	MATLAB (`matlab/`)
Bisection (Thm 1.10)	`bisection.bisection`	`bisection.m`
Φ, φ, Φ^{-1}	`normal.norm_cdf / norm_pdf / norm_ppf`	`norm_cdf.m` …
A_n, Pe^{rt}, error (3)	`compounding.compound / continuous / error_estimate`	`compound.m` …
CRR tree, Black–Scholes	`option_pricing.binomial_price / black_scholes`	`binomial_price.m` …
Walk distribution, D_n	`gbm.walk_log_return_distribution / ks_distance_to_gbm`	`ks_distance_to_gbm.m` …
g, f^*, VaR, f_{VaR}, f_{opt}	`risk.growth_rate / kelly_fraction / value_at_risk / max_fraction_under_var / optimal_fraction`	same names `.m`
Monte Carlo §5.7	`risk.simulate_wealth / empirical_var`	`simulate_wealth.m`, `quantile_linear.m`
Quadratic root (Prop 5.4), ES (Prop 5.5)	`risk.var_root_formula / expected_shortfall / max_fraction_under_es`	same names `.m`
Crashes (Part VI)	`risk.poisson_pmf / value_at_risk_jump / max_fraction_under_var_jump / growth_rate_jump / kelly_fraction_jump`	same names `.m`
Best mix and its limit (Part VII)	`portfolio.fund_fraction / best_mix_under_var / best_mix_on_grid / refine_best_mix / near_optimal_ranges`	same names `.m`

Numerical details worth knowing

- **$\log 1p$ for compounding.** For $n = 10^7$, $1 + r/n = 1.00000001$ loses digits in floating point. The code computes $\exp(nt \cdot \log 1p(r/n))$, where `log1p(u)` evaluates $\ln(1+u)$ accurately for tiny u.
- **Logarithms for binomial probabilities** (§4.3) avoid overflow of $\binom{n}{k}$.
- **No statistics libraries.** $\Phi(x) = \frac{1}{2}\left(1 + \operatorname{erf}(x/\sqrt{2})\right)$ and Φ^{-1} by bisection, so the code is self-contained and identical in Python and MATLAB.
- **Exactness vs. simulation.** Modules 1–3 and the optimisation are deterministic (Python and MATLAB agree to many digits). Only the Monte Carlo checks are random; they differ slightly between languages.

Part IX — Summary, assumptions, and likely questions

9.1 Where each theorem is used

Mathematical idea	Where
Limit of a sequence, squeeze, monotone convergence	$(1+r/n)^{nt} \to e^{rt}$ (Thm 2.1, Prop 2.2)
Continuity: limit passes inside **exp**	Thm 2.1, Thm 5.1, every exponential model
Taylor expansion / asymptotics	rate $1/n$ (§2.3), expansion of p (§3.4), Thm 5.1
Limit in distribution, CLT	binomial tree → Black–Scholes; random walk → GBM
Rates of convergence	$1/n$ (compounding), $1/N$ (options), $1/\sqrt{n}$ (KS distance)
Derivative as a limit; concavity	$g'(f)$, Kelly fraction f^*
Intermediate Value Theorem	existence of f_{VaR}, f_{ES}, $z_{0.95}$, the root of g', the crash quantile, f_{crash}, f^*_{crash}, and $f_{VaR}(d)$ for every fund
Bisection (nested intervals)	computing all these roots; 41 iterations on $[0, 2]$
Concavity, quadratic roots	uniqueness of f_{VaR} and f_{ES} for every horizon (Props 5.4, 5.5); exactly one best mix (Thm 7.2)
Extreme Value Theorem	existence of the optimal allocation on $[0, c]$ and of the best mix
Poisson limit	crash coin-flips → Poisson law (Thm 6.1)
Monotone convergence, uniform continuity	$(1+r/n)^n$ increases to its limit; finer grids of mixes increase to the best mix (Thm 7.3)
Law of large numbers	$\sum Z_k^2 \Delta t \to T$ (Thm 5.1); Monte Carlo check

9.2 Assumptions and limitations

- Constant μ, σ, r; in reality they change and μ is hard to estimate.
- Normal log-returns. Real returns have *fat tails*, so the true 95% VaR is somewhat higher than the model's.
- Free, continuous rebalancing; borrowing at the risk-free rate; no taxes.
- VaR ignores *how bad* losses beyond the 5% threshold are. The Expected Shortfall of §5.8 measures it, and the same IVT/bisection method works for it, since it is also continuous in f.
- Fat tails come only from the simple crash model of Part VI, with an assumed crash size and frequency. The gold and bond numbers of Part VII are assumptions, and the best mix has a flat top, so its exact split is not robust; “about 73% stocks” is.

9.3 Questions you may be asked

Where exactly is the limit in your project?

Six places: $(1+r/n)^{nt} \to e^{rt}$; the binomial price $C_N \to C_{BS}$; the random walk → GBM as $\Delta t \to 0$; the derivative $g'(f)$ as the limit of difference quotients; crash coin-flips → the Poisson law; and the best grid

mix $\to$ the exact best mix as the grid step $\to 0$. Theorem 5.1 (discrete rebalancing $\to$ continuous) combines the first four.

Is 0.705 the only allocation with VaR exactly 15%?

Yes, for any horizon. $\mathrm{VaR}(f) = L$ is a quadratic equation in f whose two roots have a negative product, so only one root is positive (Proposition 5.4). The quadratic formula gives 0.705014 too.

Where is continuity used, and why is it essential?

The VaR function is continuous in f, with $h(0) < 0 < h(2)$, so the IVT guarantees an allocation with risk exactly 15%. Without continuity (for example, if risk could jump) the root might not exist and bisection could fail.

Why not just solve the quadratic?

Here we can, and it confirms 0.705014. But bisection only needs continuity, so it still works for realistic risk measures that have no formula.

How do you know bisection stops after 41 steps?

The interval halves each time: after k steps its length is $2/2^k$. We need $2/2^k < 10^{-12}$, i.e. $k > \log_2(2 \cdot 10^{12}) \approx 40.9$.

Why not invest the Kelly fraction 1.75?

It needs 75% borrowed money and has a 38.4% one-year VaR. Growth is a flat hill near the top: going from 1.75 to 0.705 reduces growth only from 9.13% to 6.94%, but cuts the risk from 38.4% to 15%.

What is the $-\frac{1}{2}\sigma^2$ term?

Volatility drag: the average price grows at μ, but the typical (median, log) growth is $\mu - \frac{1}{2}\sigma^2$, because losses hurt more than equal gains help (-50% needs $+100\%$ to recover). It is also why growth falls when you take too much risk.

Why does the binomial price oscillate?

The strike's position relative to the tree's final nodes alternates with the parity of N, so the error constant alternates ($N\cdot$error ≈ 2.00 for even N, 1.75 for odd N). Both still go to zero like $1/N$.

How did you check the results are right?

35 automated tests (repeated in MATLAB) compare against known values ($e^{0.1}$, the Black–Scholes reference 10.4506, put–call parity, $z_{0.95}$, $f^* = 1.75$), check the limits (Poisson, finer grids never worse), check that every crash function equals its bell-curve twin when $\lambda = 0$, and check the Expected Shortfall. The closed-form root confirms bisection, the $1/\sqrt{n}$ prediction matches the KS distances, and Monte Carlo simulation confirms the VaR and the ES.

References

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